SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	**Renuga Devi B**	**Reg. No.:**	**192311111**
Section / Batch:		**Date:**	**30/10/2026**

Code of Conduct

*I **Renuga Devi B (192311111)** (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ***Renuga Devi B***

Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

Annexure A - QUESTIONS

1. Design a concept map that evaluates the significance of the fundamental concepts of mobile computing.
   Establish relationships among mobility, portability, wireless communication, and ubiquitous computing.
   Prioritize the concepts based on their contribution to mobile computing.
   Justify the linking phrases used between the concepts.

2. Construct a concept map that evaluates the effectiveness of the cellular network architecture.
   Illustrate the interactions among Mobile Station, Base Station Subsystem, and Network Switching Subsystem.
   Identify the most critical components influencing communication.
   Justify your evaluation through meaningful concept relationships.

3. Develop a concept map that evaluates the impact of frequency reuse on cellular network performance.

Relate frequency planning, co-channel interference, adjacent channel interference, and network capacity.
Assess the effectiveness of interference reduction techniques.
Support your evaluation using appropriate linking phrases.

4. Design a concept map that evaluates the importance of different cellular access channels.
   Illustrate how paging, broadcast, control, and traffic channels contribute to communication.
   Rank the channels based on their operational significance.
   Justify the relationships established in the concept map.

5. Construct a concept map that evaluates the contribution of Mobile Station, Base Station Subsystem, and Network Switching Subsystem toward seamless communication.
   Illustrate their interdependencies.
   Assess which subsystem has the greatest influence on service continuity.
   Justify your conclusions using concept relationships.

6. Develop a concept map that evaluates the effectiveness of the mobile network protocol stack.
   Illustrate interactions among protocol layers.
   Assess how each layer contributes to reliable mobile communication.
   Support your evaluation with meaningful hierarchical relationships.

7. Design a concept map that evaluates the role of the core network in delivering mobile communication services.
   Relate authentication, mobility management, subscriber databases, and call routing.
   Determine the most influential core network functions.
   Justify your evaluation through concept linkages.

8. Construct a concept map that evaluates the operational efficiency of a Public Land Mobile Network (PLMN).
   Relate operators, subscribers, switching centers, and radio access networks.
   Assess how these components collectively provide seamless connectivity.
   Justify the relationships included in the concept map.

9. Develop a concept map that evaluates different mobility management functions.
   Illustrate relationships among location update, registration, paging, and tracking.
   Determine which functions most effectively maintain user connectivity.
   Support your evaluation with appropriate concept links.

10. Design a concept map that evaluates various handoff mechanisms used in cellular networks.
    Compare hard handoff, soft handoff, and handoff decision parameters.
    Assess their suitability for maintaining uninterrupted communication.
    Justify your evaluation using interconnected concepts.

11. Construct a concept map that evaluates the effectiveness of roaming mechanisms in mobile communication.
    Illustrate relationships among home network, visited network, authentication, registration, and billing.
    Assess the importance of each process in successful roaming.
    Support your conclusions through logical concept organization.

12. Develop a concept map that evaluates the relationship between mobility, handoff, and roaming.
    Illustrate their dependencies and interactions.
    Determine which mechanism contributes most to uninterrupted communication.
    Justify your evaluation using appropriate linking phrases.

13. Design a concept map that evaluates the influence of cellular performance criteria.
    Relate coverage, capacity, Quality of Service (QoS), reliability, throughput, and delay.
    Assess which criteria most significantly affect user experience.
    Justify the hierarchy established in the concept map.

14. Construct a concept map that evaluates the effectiveness of various cellular system operations.
    Include call setup, authentication, paging, resource allocation, handoff, and call termination.
    Assess the contribution of each operation to overall network performance.
    Support your evaluation through logical concept relationships.

15. Develop a concept map that evaluates the complete mobile communication process.
    Illustrate the interaction among user equipment, access network, core network, mobility management, and service delivery.
    Determine the most critical stages in ensuring communication quality.
    Justify your evaluation with meaningful linking phrases.

16. Design a concept map that evaluates the relationship between frequency reuse, interference, and network capacity.
    Assess how each factor influences cellular system efficiency.
    Prioritize the concepts based on their impact.
    Justify the relationships shown in the concept map.

17. Construct a concept map that evaluates the interdependence among cellular network architecture, protocol stack, and mobility management.
    Assess how these components collectively improve communication reliability.
    Identify the strongest conceptual relationships.
    Justify your evaluation using appropriate linkages.

18. Develop a concept map that evaluates resource management strategies in cellular networks.
    Relate channel allocation, power control, interference management, frequency reuse, and handoff management.
    Assess which strategy contributes most to efficient network utilization.
    Support your evaluation through concept relationships.

19. Design a concept map that evaluates the integration of all major concepts in mobile computing.
    Include architecture, protocol stack, mobility management, roaming, handoff, performance criteria, and cellular operations.
    Assess how these concepts collectively support an efficient mobile communication system.
    Justify the hierarchy and linking phrases used.

20. Construct a concept map that evaluates alternative approaches for improving cellular network performance.
    Relate mobility management, handoff optimization, interference reduction, protocol efficiency, and resource allocation.
    Recommend the most effective combination of techniques.
    Justify your recommendation through well-defined conceptual relationships.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Fundamental Concepts of Mobile Computing

Introduction

Mobile computing is a technology that allows users to access information, communicate, and use applications while moving from one place to another. It mainly depends on mobility, portability, wireless communication, and ubiquitous computing.

1. Mobility

Mobility allows users to communicate and access network services while moving between different locations. For example, a person can make a phone call or access the internet while travelling.

2. Portability

Portability refers to the ability to carry mobile devices easily from one place to another. Smartphones, tablets, and laptops are examples of portable devices. It makes mobile computing convenient for everyday use.

3. Wireless Communication

Wireless communication allows devices to exchange information without physical cables. Technologies such as Wi-Fi, Bluetooth, and cellular networks help mobile devices connect and communicate with other devices and networks.

4. Ubiquitous Computing

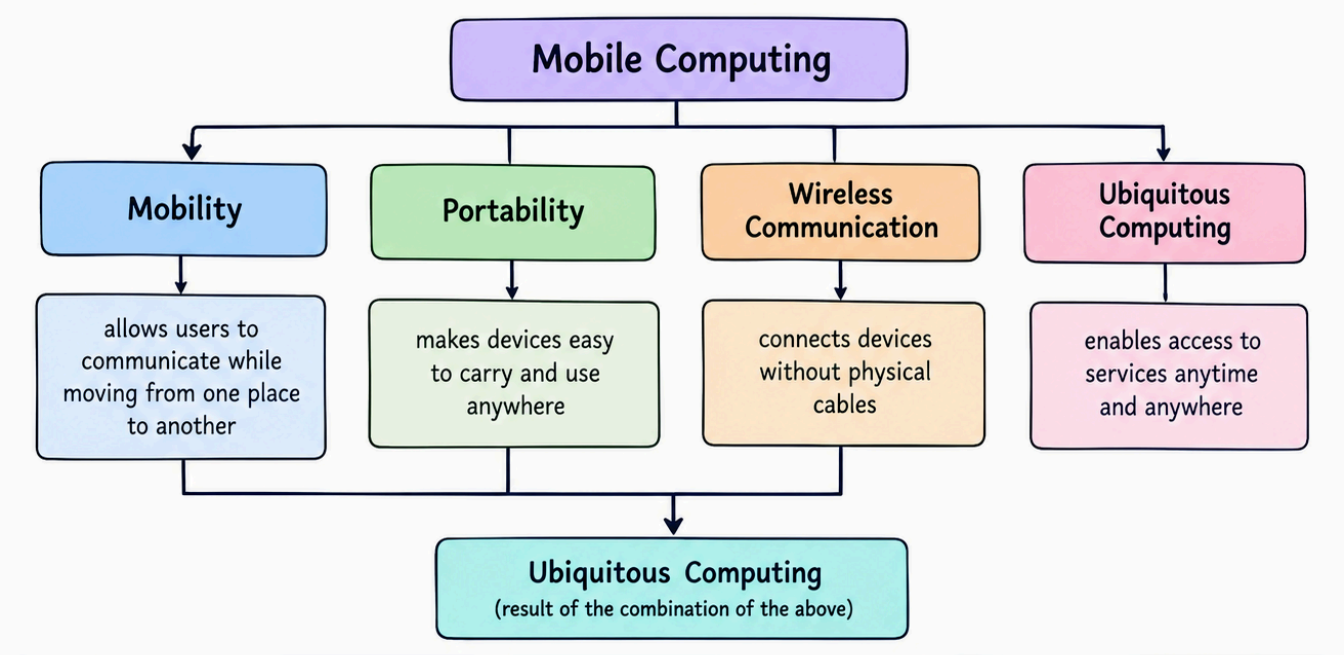
Ubiquitous computing refers to the availability of computing services almost anytime and anywhere. It allows users to access information and applications whenever they need them, provided that the required network and services are available.

Evaluation and Priority

Wireless communication and mobility are the main foundations of mobile computing because they support connectivity and communication while moving. Portability makes devices easier to carry and use. Ubiquitous computing represents the broader goal of making computing services readily accessible in different locations.

Conclusion

Mobility, portability, wireless communication, and ubiquitous computing are closely related concepts. Together, they make mobile computing useful, convenient, and accessible in everyday life.



Concept map of Fundamental Concepts of Mobile Computing

1. Evaluation / Justification

Mobility allows users to communicate while moving from one place to another. Portability makes mobile devices easy to carry and use. Wireless communication connects devices without physical cables and supports data transfer. Ubiquitous computing provides access to computing services anytime and anywhere, depending on network availability. All four concepts contribute to making mobile computing useful and convenient.

2. Priority

1. Wireless Communication: It is essential for connecting mobile devices to networks.
2. Mobility: It allows users to communicate while moving between locations.
3. Portability: It makes devices convenient to carry and use.
4. Ubiquitous Computing: It represents the broader goal of providing computing services anytime and anywhere.

3. Justification of Linking Phrases

- "Allows users to communicate while moving" explains how mobility supports communication across different locations.
- "Makes devices easy to carry and use" describes the importance of portability in mobile computing.
- "Connects devices without physical cables" explains how wireless communication enables information exchange.
- "Enables access to services anytime and anywhere" shows the purpose of ubiquitous computing.
- "Together they support mobile computing" indicates that all four concepts contribute to the effective functioning of mobile computing.

2. Effectiveness of Cellular Network Architecture

Introduction

Cellular network architecture is the structure of a mobile communication system that allows users to make calls, send messages, and access data services. It consists of three main components: Mobile Station (MS), Base Station Subsystem (BSS), and Network Switching Subsystem (NSS). Each component performs a specific function, and all three work together to provide effective cellular communication.

1. Mobile Station (MS)

The Mobile Station refers to the mobile device used by the user to make and receive calls, send messages, and access the internet. It communicates with the base station through radio signals. Smartphones are common examples of mobile stations.

2. Base Station Subsystem (BSS)

The Base Station Subsystem provides the radio connection between the mobile device and the cellular network. In a traditional GSM network, it consists of the Base Transceiver Station (BTS) and Base Station Controller (BSC). It helps maintain communication between mobile devices and the network.

3. Network Switching Subsystem (NSS)

The Network Switching Subsystem manages call routing, subscriber information, and mobility between network areas. It connects calls to their destinations and helps users maintain communication when moving between different locations. In traditional GSM networks, it includes components such as the Mobile Switching Centre (MSC), Home Location Register (HLR), and Visitor Location Register (VLR).

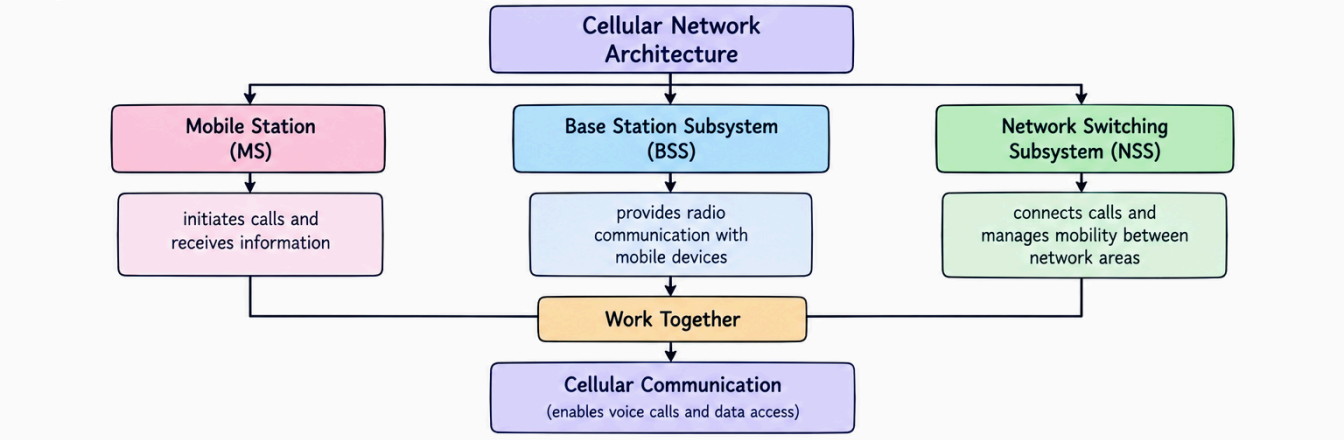
Evaluation and Priority

All three components are essential for effective cellular communication. The Base Station Subsystem is important because it establishes the radio connection between the mobile device and the network. The Network Switching Subsystem is necessary for call routing and mobility management, while the Mobile Station allows users to initiate and receive communication.

The BSS can be given priority for establishing the radio link, followed by the NSS for call routing and mobility management, and the MS for user interaction. However, the actual importance of each component depends on the communication task.

Conclusion

The Mobile Station, Base Station Subsystem, and Network Switching Subsystem work together to provide cellular communication. Their proper coordination enables users to make calls, exchange information, and access data services effectively. Therefore, the effectiveness of cellular network architecture depends on the reliable operation of all three components.



Concept map of Effectiveness of Cellular Network Architecture

1. Evaluation / Justification

The cellular network architecture consists of three important components: Mobile Station (MS), Base Station Subsystem (BSS), and Network Switching Subsystem (NSS). The MS allows users to make and receive calls. The BSS provides the radio connection between the mobile device and the network. The NSS connects calls to their destinations and manages mobility between network areas. All three components work together to provide effective cellular communication. If any component fails, communication may be interrupted.

2. Priority

1. Base Station Subsystem (BSS): It establishes the radio connection between the mobile device and the network.
2. Network Switching Subsystem (NSS): It connects calls to their destinations and supports mobility management.
3. Mobile Station (MS): It allows users to initiate and receive calls and access data services.

The BSS is given the highest priority for establishing the radio link, while the NSS is essential for call routing and mobility management. All three components are necessary for proper network operation.

3. Justification of Linking Phrases

- "Initiates calls and receives information" explains the role of the Mobile Station in user communication.
- "Provides radio communication" describes how the Base Station Subsystem connects mobile devices to the network.
- "Connects calls and manages mobility" explains the role of the Network Switching Subsystem in call routing and movement between network areas.
- "Work together" shows the interdependence of all three components.
- "Enables voice calls and data access" describes the final result of their combined operation.

3. Impact of Frequency Reuse on Cellular Network Performance

Introduction

Frequency reuse is a technique used in cellular networks to increase network capacity by using the same frequency channels in different cells that are sufficiently separated. It helps serve more users using the limited available frequency spectrum. However, improper frequency planning can cause interference and reduce communication quality.

1. Frequency Planning

Frequency planning is the process of assigning frequency channels to different cells in a cellular network. Proper planning reduces interference between nearby cells and improves signal quality. It helps balance network capacity and communication performance.

2. Co-channel Interference

Co-channel interference occurs when two or more cells use the same frequency and their signals interfere with each other. It can reduce signal quality and affect call clarity and data transmission. Proper cell separation and frequency reuse planning help reduce this interference.

3. Adjacent-channel Interference

Adjacent-channel interference occurs when signals from nearby frequency channels interfere with each other. It may happen because of imperfect filtering or insufficient frequency separation. Proper frequency allocation and filtering techniques help minimize this problem.

4. Interference Reduction Techniques

Interference reduction techniques improve cellular network performance by reducing unwanted signal interference. These techniques include proper frequency planning, suitable cell spacing, power control, and improved filtering. They help maintain good signal quality while allowing efficient frequency reuse.

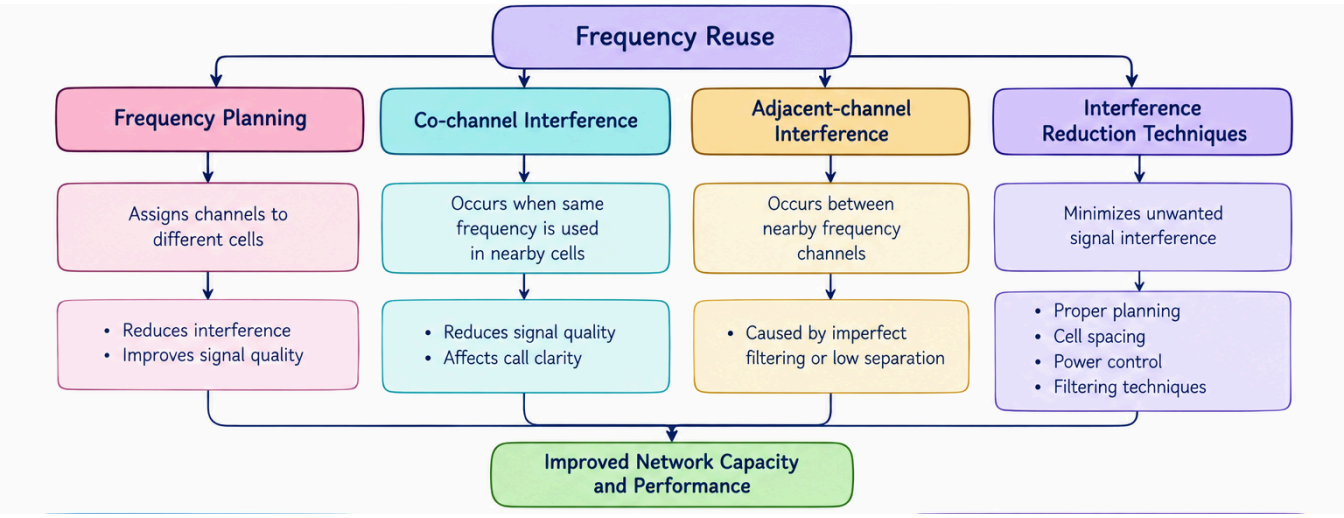
Evaluation and Priority

Frequency reuse is important because it increases network capacity without requiring additional frequency spectrum. However, excessive frequency reuse may increase co-channel interference, while poor frequency allocation can cause adjacent-channel interference.

Proper frequency planning should be given the highest priority because it helps prevent interference. Interference reduction techniques are also essential for maintaining signal quality. Therefore, an effective cellular network must balance frequency reuse, network capacity, and interference control.

Conclusion

Frequency reuse improves cellular network capacity by allowing the same frequencies to be used in different cells. However, its effectiveness depends on proper frequency planning and interference reduction. By controlling co-channel and adjacent-channel interference, cellular networks can provide better signal quality and more reliable communication.



Concept map of Frequency Reuse on Cellular Network Performance

1. Evaluation / Justification

Frequency reuse increases network capacity by allowing the same frequencies to be used in different cells. However, it may cause co-channel interference when nearby cells use the same frequency. Adjacent-channel interference occurs when signals from nearby frequency channels interfere with each other. Proper frequency planning, suitable cell spacing, power control, and filtering techniques help reduce interference. Therefore, frequency reuse improves network performance when interference is controlled effectively.

2. Priority

1. Frequency Planning: It has the highest priority because proper frequency allocation helps prevent interference between cells.
2. Interference Reduction Techniques: These techniques help maintain signal quality and improve communication reliability.
3. Co-channel and Adjacent-channel Interference Control: Reducing both types of interference improves call clarity and data transmission.
4. Frequency Reuse: It increases network capacity by allowing the same frequencies to be reused in different cells.

3. Justification of Linking Phrases

- "Assigns channels to different cells" explains the role of frequency planning in allocating frequencies.
- "Occurs when the same frequency is used in nearby cells" describes the cause of co-channel interference.
- "Occurs between nearby frequency channels" explains adjacent-channel interference.
- "Minimizes unwanted signal interference" describes the purpose of interference reduction techniques.
- "Improves network capacity and performance" shows the final benefit of proper frequency reuse and interference management.

4. Importance of Different Cellular Access Channels

Introduction

Cellular channels are used in mobile networks to establish and maintain communication between mobile devices and the network. Different channels perform different functions, such as broadcasting system information, locating mobile devices, managing call setup, and carrying voice or data. Paging, broadcast, control, and traffic channels work together to provide effective cellular communication.

1. Broadcast Channels

Broadcast channels transmit important system information from the network to mobile devices. This information helps mobile devices identify the network and understand how to access its services. They are important for enabling devices to connect to the cellular network.

2. Paging Channels

Paging channels notify mobile devices about incoming calls or other network-related activities. When a call arrives, the network uses paging to locate the intended mobile device. This helps the network reach devices that are not currently involved in an active call.

3. Control Channels

Control channels carry signalling information required to establish, manage, and release connections. They coordinate communication between mobile devices and the network. They are essential for call setup, channel allocation, and other network operations.

4. Traffic Channels

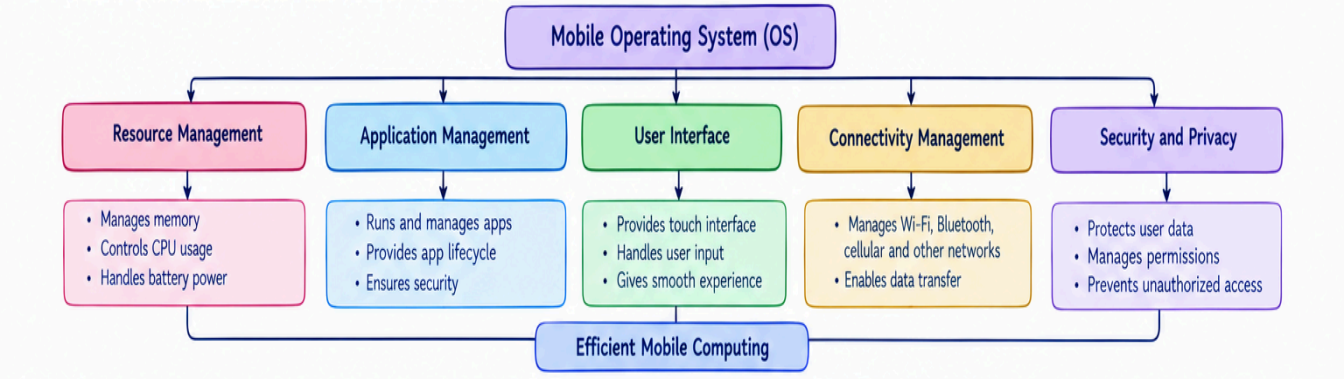
Traffic channels carry actual user information, including voice and data, during an active communication session. They allow users to conduct phone calls and exchange information over the cellular network. Without available traffic resources, normal voice or data communication cannot be carried.

Evaluation and Priority

All four types of channels are important for effective cellular communication. Broadcast channels provide system information, while paging channels help locate mobile devices for incoming calls. Control channels manage signalling and connection establishment, whereas traffic channels carry actual user information.Traffic channels can be given the highest priority during an active call because they carry voice and data. Control channels are essential for managing connections, while paging and broadcast channels support device discovery and network access.

Conclusion

Broadcast, paging, control, and traffic channels perform different but related functions in cellular networks. Their coordination helps mobile devices access network services, establish connections, receive incoming calls, and exchange information. Therefore, the proper operation of these channels is necessary for reliable cellular communication.



Concept map of Importance of Different Cellular Access Channels

1. Evaluation / Justification

Cellular channels perform different functions to support effective communication. Broadcast channels provide system information, while paging channels notify mobile devices about incoming calls. Control channels manage signalling and connection establishment, whereas traffic channels carry voice and data during active communication. All these channels work together to establish and maintain communication. Therefore, their proper operation is essential for reliable cellular network performance.

2. Priority

1. Traffic Channels: They carry actual voice and data during active communication.
2. Control Channels: They manage signalling, call setup, and connection-related operations.
3. Paging Channels: They help the network locate mobile devices for incoming calls.
4. Broadcast Channels: They provide system information required for network access.

This ranking is based on their roles during an active communication session. However, paging and broadcast channels are also essential before a call can be established.

3. Justification of Linking Phrases

- "Provides system information" explains the role of broadcast channels in helping mobile devices access the network.
- "Notifies mobile devices about incoming calls" describes the function of paging channels.
- "Manages signalling and connection establishment" explains how control channels coordinate communication.
- "Carries actual voice and data" shows the importance of traffic channels during an active session.
- "Supports cellular communication" explains how all four channel types contribute to effective network operation.

5. Contribution of MS, BSS, and NSS Toward Seamless Communication

Introduction

Seamless communication refers to maintaining a continuous connection with minimum interruption while a user communicates or moves between different cellular coverage areas. It depends on three important components: Mobile Station (MS), Base Station Subsystem (BSS), and Network Switching Subsystem (NSS). These components work together to maintain connectivity, manage calls, and support communication during movement between cells.

1. Mobile Station (MS)

The Mobile Station is the mobile device used by the user to make and receive calls, send messages, and access data services. It communicates with the cellular network through radio signals. It also participates in handover procedures by measuring and reporting radio signal conditions, helping maintain communication when the user moves between cells.

2. Base Station Subsystem (BSS)

The Base Station Subsystem manages the radio communication between the mobile device and the cellular network. In a traditional GSM network, it consists of the Base Transceiver Station (BTS) and Base Station Controller (BSC). It manages radio resources and supports handover procedures within its area of responsibility.

3. Network Switching Subsystem (NSS)

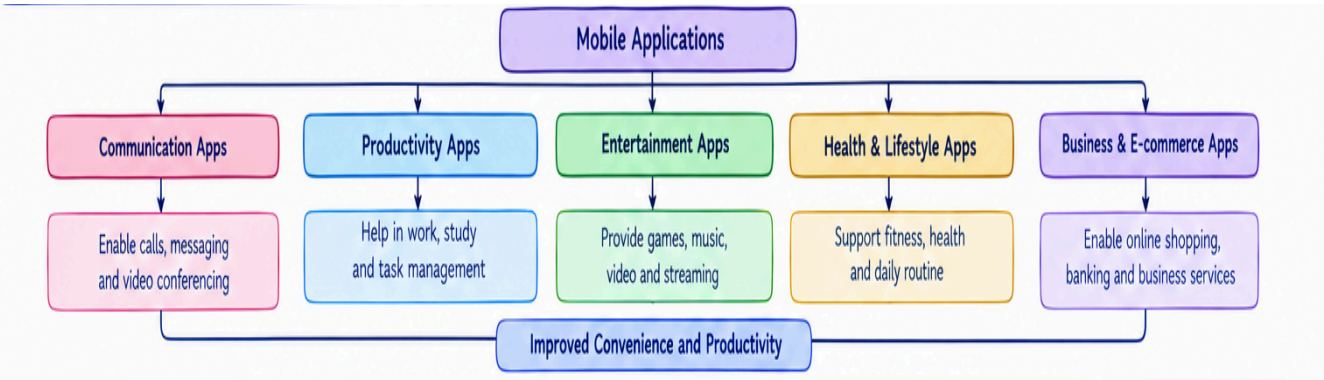
The Network Switching Subsystem manages call routing, subscriber information, and mobility between network areas. In a traditional GSM network, it includes components such as the Mobile Switching Centre (MSC), Home Location Register (HLR), and Visitor Location Register (VLR). It coordinates call connections and supports mobility management, helping users maintain communication when moving between different network areas.

Evaluation and Priority

The MS, BSS, and NSS are interdependent because each performs a different function in maintaining seamless communication. The MS enables user communication, the BSS maintains the radio connection, and the NSS manages call routing and mobility across network areas. The failure of any one component can interrupt communication.The NSS can be considered the most influential subsystem for service continuity across different network areas because it supports call routing and mobility coordination.

Conclusion

The Mobile Station, Base Station Subsystem, and Network Switching Subsystem contribute to seamless communication through user connectivity, radio resource management, call routing, and mobility coordination. Their interdependence helps reduce communication interruptions when users move between cells. Although the NSS has a major influence on network-wide service continuity, all three components are necessary for reliable and continuous mobile communication.



Concept map of Contribution of MS, BSS, and NSS Toward Seamless Communication

1. Evaluation / Justification

Seamless communication depends on the coordination of the Mobile Station (MS), Base Station Subsystem (BSS), and Network Switching Subsystem (NSS). The MS allows users to make and receive calls, while the BSS maintains the radio connection between mobile devices and the network. The NSS manages call routing and mobility across network areas. Their combined operation helps reduce communication interruptions when users move between cells. Therefore, all three components are essential for maintaining service continuity.

2. Priority

1. Network Switching Subsystem (NSS): It supports call routing and mobility coordination across network areas.
2. Base Station Subsystem (BSS): It maintains radio connectivity and supports handover between cells.
3. Mobile Station (MS): It enables user communication and participates in mobility procedures.

The NSS is given the highest priority for network-wide coordination, while the BSS is critical for maintaining the radio connection during movement. However, seamless communication requires all three components to function correctly.

3. Justification of Linking Phrases

- "Initiates and receives calls" explains the role of the Mobile Station in user communication.
- "Maintains the radio connection" describes how the Base Station Subsystem connects mobile devices to the network.
- "Routes calls and supports mobility management" explains the role of the Network Switching Subsystem in maintaining communication across network areas.
- "Work together to provide seamless communication" shows the interdependence of all three components.
- "Maintains communication when users move between cells" explains the importance of handover and network coordination in reducing interruptions.